

SmartChef AI – Recipe Preparation Agent

Problem Statement

Many people struggle to decide what to cook using only the ingredients already available in their kitchen. Most recipe applications recommend meals that require additional ingredients, leading to unnecessary grocery purchases, higher costs, and increased food waste. There is a need for an intelligent assistant that can suggest personalized recipes based on available ingredients while considering dietary preferences, allergies, cooking time, and nutritional requirements.

Proposed Solution

SmartChef AI is an AI-powered Recipe Preparation Agent developed using IBM Granite, IBM watsonx.ai, IBM watsonx Orchestrate, and Retrieval-Augmented Generation (RAG). The agent retrieves relevant recipes from a structured knowledge base and generates personalized cooking instructions, ingredient substitutions, nutritional insights, and cooking tips through a conversational interface. The solution helps users prepare meals efficiently, reduce food waste, and improve the overall cooking experience.

Project Objectives

- Recommend recipes using available ingredients.
- Reduce household food waste.
- Support dietary preferences and allergies.
- Provide step-by-step cooking guidance.
- Suggest ingredient substitutions.
- Deliver a personalized AI-powered cooking experience using IBM Granite and IBM watsonx Orchestrate.